

DEV@Deakin App

To achieve a HD in this unit, you must demonstrate mastery over the SIT313 topics covered, and also your own creativity and interests in frontend application development.

Complete the following:

1. Complete the Task Requirements.
2. Create a new [GitLab](#) repository for this task. Ensure all files/resources created are committed to this repository.
3. Create a detailed report that includes:
 - a. Screenshots of your DEV@Deakin web platform.
 - b. Descriptions and screenshots of each addition/modification made to your DEV@Deakin web platform, with explanations of why you chose to implement each, and reflections on challenges faced and lessons learned.
 - c. Justification for where and how your solution is at a HD-level.
 - d. Screenshot of your seminar/HelpHub tutors' email confirmation that you have had discussions with them during the trimester about your solution/ideas. Your submission will not be accepted if it is missing this criteria.
 - e. A link to your repository on GitLab, with regular commits (with proper comments) made over a period of weeks.
4. **Upload** to OnTrack and **engage with feedback** to get this complete!

Task Requirements

Read the below instructions clearly to ensure your solution is at the right level of complexity and scope.

Note that in addition to completing the below requirements, you must also attend a live demonstration with the SIT313 teaching team at trimester's end, as discussed below.

1. GitLab repository

- Create a New Project called Task HD1. Set the Visibility Level to **Private**.
- Once the project is created, under **Manage->Members** add the Unit Chair (Aaron Spence) and your OnTrack marking tutor **as a Reporter role**.
- Strictly do not submit your `node_modules` folder to your repository. To prevent this, create a `.gitignore` file, and add inside it `node_modules/` to prevent this folder from being tracked by Git.
- Commit all code and resources (e.g. images) that you develop for this task into this repository, with proper commit comments.

Failure to follow the above steps means your task cannot be marked as Complete.

2. Set Your Own Direction

We have spent the trimester developing DEV@Deakin to what it currently is. You are now free to decide how and where you wish to expand it, depending on your own areas of interest.

You have the option to develop one large feature, or many smaller features. At a minimum your solution must:

- Be a continuation of your DEV@Deakin web platform, using a React frontend, a NodeJS backend (either ExpressJS or NextJS), and have *all* previous features developed from the Pass, Credit and Distinction tasks.
- Have additions/modifications to DEV@Deakin that are beyond what has been covered in any SIT313 Pass, Credit, or Distinction task, in terms of complexity and polish. Includes use of more React hooks, such as `useMemo`, `useContext`, and/or `useOptimistic`.

- Used industry best-practices: TypeScript, separation between frontend and backend with RESTful practices, robust data validation, proper Git commits with comments at regular intervals, extremely well-structured code with comments, and all features implemented to a high standard.

3. Receiving Feedback

As part of completing this task, it is a core requirement that you have discussions with the SIT313 teaching team about your submission ideas at various points along the trimester (either on-campus or online – discussions over email are not permitted). Seek feedback to verify:

- Scope – A very large, overly complex project does not guarantee acceptance, nor would a solution that is too small.
- Complexity - We are looking for quality over quantity. Your submission should be more complex than what has been covered in SIT313, but also not so complex that it takes months to complete, or is too large for us to properly discuss with you and assess.

Before submitting this task, request from the tutors you have been discussing with an email confirming your discussions about your ideas and progress. Take a screenshot of this email to include in your report as evidence.

Important Notes

- Your submission will strictly not be accepted if it is missing this criteria. We suggest having discussions early, and ideally over multiple instances to get continued feedback.
- You cannot have discussions with your tutors about your ideas over email or Teams – it must occur in-person (either on-campus or online) during a seminar or HelpHub.

4. Prepare for the Live Demonstration

Once you submit this task, you may be eligible to attend the required live demonstration with the SIT313 teaching team. These will occur online via Teams during Weeks 11 and 12 and last 10-15 minutes - information about how to join will be shared in the SIT313 Teams channel around Week 9.

To be successful (and thus receive a Complete for this task), we are *not just assessing the quality of your code*. We will also crucially be assessing your *understanding* of your developed solution.

The live demonstration is not meant to be stressful or be considered as an interrogation of course! We will be asking about:

- Why you made certain design decisions?
- Why different technologies were chosen and how exactly were they implemented?
- What happens when we ask you to demonstrate test cases to see if your developed features work as intended? Can you explain the outcome?

Please ensure that if you are signing up for a live demonstration that your submission is already to a very high standard and is deserving of a High Distinction grade, and you have a deep understanding of your own solution.

Caution: Acceptable use of Artificial Intelligence

It is acceptable to use AI as a learning tool. However, you can **never** ask AI to write code to complete some/parts of this task for you: this is cheating because it does not demonstrate that you understand anything. Detection of inappropriate use of AI has a very real consequences, such as failing the task, or even failing the entire unit.

Sample Ideas

Here are some ideas of what you may implement for this task (although ideally you can think of your own too):

- Two-Factor Authentication (2FA) for secure user management.
- Lazy loading and pagination to handle large volumes of user submitted Questions/Articles.
- A much more feature-rich Gallery.
- Administration dashboards to control and monitor different functionality across the platform.
- Additional features for Paid users.
- A trial period for a Free user to trial a Paid account for a period of time.

How Do I Know My Solution Deserves a HD grade?

These above sample ideas are simply to help you get started - implementing them is not a requirement. Implementing your own ideas that are the same or similar would be a good starting point, but you must understand that it is not the ideas you implement into your website that is important - it is the *quality* on how they are implemented.

Furthermore, it is also how well you can demonstrate them during the live demonstration.

Best is to have regular discussions with the SIT313 teaching team during your seminars, to receive feedback on your ideas and your progress.

Marking Guide

Your success in this task is not just based on your submitted code – it is the culmination of the following areas:

Competent	Proficient
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Report	
Report provides a general idea of the intended features. Each feature also has a paragraph explaining the feature in adequate detail. Overall presentation is decent.	Report provides clear and concise explanations of the intended features. Each feature includes details of how that feature operates, with reflections on how it was implemented and why. Presentation and structure is excellent.
Documentation	
Commenting is present throughout and provides acceptable readability of the code for someone unfamiliar with the project. Files are well structured.	Commenting is present, thorough and sophisticated throughout without overwhelming the project. The code is very easily readable for a user unfamiliar with the project. Consistency is highly maintained.
DEV@Deakin	
Satisfies coding requirements effectively with creative or interesting functionality, including at least three new advanced React concepts. Used effectively as a part of the larger system with logical implementation and fully functional DEV@Deakin. Evidence of implementation beyond what has been demonstrated in class.	Exceeds advanced coding requirements effectively and/or demonstrates highly creative or interesting functionality including at least three new advanced React concepts. Sophisticated integration demonstrates thorough understanding of the concept. Clear evidence of functionality beyond what has been demonstrated in class and a fully functional DEV@Deakin app.
Presentation	
You demonstrate part of features as working, and can explain each with adequate detail and understanding. It is obvious you invested thought and time into your solution.	You demonstrate all main features as working, and can explain each with ease and in-depth. It is obvious you invested considerable thought and time into your solution.